AI-Driven Prior Authorization Automation for Healthcare

Kavya K

Walsh College

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Keya Choudhury

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Abstract

Prior Authorization (PA) is a critical but inefficient administrative process in healthcare that often leads to treatment delays, increased operational costs, and provider dissatisfaction. *(Allen & Saba, 2023; Lenert, Lane, & Wehbe, 2023)*. This study proposes an AI-driven framework leveraging machine learning (ML) and natural language processing (NLP) to automate and optimize the prior authorization process. The research focuses on predicting authorization outcomes, identifying key influencing factors, extracting relevant clinical information from unstructured data, and improving operational efficiency. This study applies regression, classification, and NLP techniques *(McKinsey & Company, 2022; Aiwerioghene & Osuchukwu, 2025)*. The findings aim to demonstrate improvements in turnaround time, approval rates, and decision-making efficiency, contributing to enhanced healthcare delivery and reduced administrative burden

Introduction

Background and Context

The healthcare industry is increasingly adopting digital systems such as Electronic Medical Records (EMR) to streamline clinical and administrative workflows. Despite these advancements, the Prior Authorization (PA) process remains a significant bottleneck *(Allen & Saba, 2023)*. PA requires healthcare providers to obtain approval from insurance payers before delivering certain treatments, procedures, or medications. While intended to control costs and ensure appropriate care, the process is largely manual and inefficient *(Lenert et al., 2023)*

Healthcare systems such as EPIC and Cerner integrate PA workflows; however, these systems still rely heavily on manual documentation, rule-based checks, and human intervention. This leads to delays in treatment, increased administrative workload, and potential negative patient outcomes *(HCLTech, 2023)*. According to industry reports, physicians spend several hours weekly on PA-related tasks, contributing to burnout and inefficiencies *(McKinsey & Company, 2022)*

Artificial Intelligence (AI), particularly machine learning and natural language processing, offers promising solutions to automate decision-making and reduce manual effort *(Frontiers Editorial Team, 2025)*.

Problem Statement

The current prior authorization process is inefficient, inconsistent, and resource-intensive *(Allen & Saba, 2023)*. It involves manual data entry, fragmented communication between providers and payers, and reliance on incomplete or unstructured clinical documentation. These challenges result in delayed approvals, high denial rates, and increased operational costs *(Lenert et al., 2023)*. There is a lack of intelligent systems capable of predicting authorization outcomes and assisting providers in preparing complete and accurate submissions. This gap leads to suboptimal decision-making and inefficiencies in healthcare delivery. Therefore, there is a need for an AI-driven solution to automate and optimize the prior authorization process *(Aiwerioghene & Osuchukwu, 2025)*.

Purpose of the Study

The primary objective of this study is to design and evaluate an AI-based system for prior authorization automation. Specifically, the study aims to:

- Identify clinical and demographic factors associated with the use of high-cost, prior-authorization-prone services.
- Develop machine learning models to predict which synthetic patients are likely to utilize such services.
- Apply NLP methods to extract key clinical information from EHR-like clinical text that is relevant to PA decision-making.
- Use real-world plan-level PA metrics to estimate how AI-based automation could affect approval patterns and turnaround time

Scope

This study focuses on the intersection of prior authorization decision support and AI using a multi-source, fully public data strategy. Structured patient- and claim-level variables are obtained from CMS Medicare Claims Synthetic Public Use Files *(Centers for Medicare & Medicaid Services, 2026)* and a de-identified EHR-like ICU dataset from Kaggle *(Johnson et al., 2024)*, while real-world prior authorization performance is characterized using the Washington Office of the Insurance Commissioner (OIC) Health Plan Prior Authorization dataset *(Data.Healthcare.gov, 2026)*. The analysis integrates structured and unstructured data, including

demographics, chronic conditions, hospital encounter characteristics, and clinical text, to model PA-prone utilization and to explore potential efficiency gains from automation.

Research Questions

- **RQ1:** ***Explainability / Factor Analysis*** *(Allen & Saba, 2023)* **-** What clinical and demographic factors significantly influence prior authorization outcomes?
    - ***H_0:*** Clinical and demographic factors have no significant influence on prior authorization outcomes.
    - ***H_1:*** Certain factors (e.g., age, diagnosis, insurance type) significantly influence outcomes
- ***RQ2: Predictability / Classification*** *(McKinsey & Company, 2022)* **-** Can machine learning models accurately predict approval versus denial?
    - **H_0:** Machine learning models cannot predict approval vs. denial better than random chance.
    - **H_1**: ML models achieve prediction accuracy significantly above chance (>50%)
- ***RQ3: Information Extraction / NLP Classification*** *(Hu et al., 2025; Pindi, 2024)* Can natural language processing effectively extract key clinical information from unstructured notes?

- **H_0:** NLP techniques (NER, TF-IDF) do not extract relevant clinical information with accuracy above baseline.
- **H_1:** NLP techniques can extract key information with statistically significant precision/recall.

- ***RQ4: Efficiency Analysis / Simulation*** *(Aiwerioghene & Osuchukwu, 2025; Frontiers Editorial Team, 2025)* **-** How does AI-based automation impact turnaround time and operational efficiency?
  - **H_0:** AI-based automation does not reduce turnaround time compared to manual processes.
  - **H_1:** AI-based automation significantly reduces turnaround time and improves efficiency

Table 1 : Sample size calculation

Research Question	**Method Used**	**Key Parameters**	**Minimum Sample Size (N)**
RQ1	Logistic regression / Chi-square	$\alpha = 0.05$, Power = 0.8, Medium effect size (Cohen's $w \approx 0.3$)	~80–100 per group (≈369 for CI with ±5% margin)
RQ2	Binary classification	10–20 samples per predictor;	~200+ (rule of thumb) or

Research Question	Method Used	Key Parameters	Minimum Sample Size (N)
	(Logistic regression, Decision Trees)	Accuracy CI ±5%	≈288 for CI
RQ3	Named Entity Recognition (NER)	Precision/Recall CI ±5%; annotated documents	~200–300 training docs + 50–100 test docs (≈246 entities for CI)
RQ4	Two-sample t-test (Before vs After)	$\alpha = 0.05$, Power = 0.8, Medium effect size (Cohen's d ≈ 0.5)	~64 per group (≈35 for CI with $\sigma = 3$ days, margin ±1 day)

Note. CI = Confidence Interval; α = significance level; Power = probability of correctly rejecting null hypothesis.

To ensure adequate statistical power across all planned analyses, the final sample size was determined by selecting the maximum of the minimum sample sizes calculated for each research question. Based on the power of test and confidence interval approaches, the required minimum sample sizes were approximately 369 for RQ1, 288 for RQ2, 246 for RQ3, and 64 for RQ4. Since RQ1 required the largest sample size (369), this value was adopted as the final sample size for the study. Using 369 participants/records ensures that all research questions can

be addressed with sufficient statistical rigor, precision, and reliability, while avoiding underpowered analyses in any domain.

Hence the study should therefore use a minimum sample size of 369 to ensure adequate statistical power across all research questions. This way, dataset will be sufficiently robust to handle all planned analyses without underpowering any of them.

Data Description

Claims data: CMS Medicare SynPUFs

Structured claims-like data will be drawn from the Medicare Claims Synthetic Public Use Files (SynPUFs) released by the Centers for Medicare & Medicaid Services (CMS). The DE1_0_2008 Beneficiary Summary file provides beneficiary-level demographics (date of birth, sex, race, state, county), coverage history (months of Part A, Part B, and HMO coverage), and chronic condition flags (e.g., diabetes, heart failure, chronic kidney disease), along with annual reimbursement totals for inpatient, outpatient, and carrier services. Linked SynPUF claims files supply diagnosis and procedure codes (ICD, HCPCS/CPT) and allowed amounts at the claim level. These data will be used to construct features such as age, sex, race, comorbidity profile, coverage history, cost burden, and binary flags for “prior-authorization-prone” services based on empirically selected procedure codes.

EHR-like ICU encounter data: Kaggle dataset

To approximate electronic health record structure and provide EHR-like clinical features and text, the study uses a public ICU dataset from Kaggle containing de-identified unit-stay-level records. Each record corresponds to an ICU stay and includes patient demographics (age, gender, ethnicity), hospital context (hospital and ward identifiers, ICU unit type, admission source), admission diagnosis (apacheadmissiondx), anthropometrics (height and weight), and discharge outcomes (discharge location and status, including disposition to home or skilled nursing facility). The dataset also provides a free-text field (clinical_text) summarizing key attributes of the admission and a search_query label that describes the note context (e.g., "heart disease diagnosis," "hospital admission details"). These structured variables will be used as additional predictors in patient- and encounter-level models, while the clinical_text field will serve as input to NLP methods in RQ3.

Plan-level PA metrics: Washington OIC Health Plan Prior Authorization Data

Real-world prior authorization patterns will be characterized using the Health Plan Prior Authorization Data published by the Washington Office of the Insurance Commissioner. Under RCW 48.43.0161, health carriers with at least 1% market share in Washington must report annual, aggregated, de-identified PA data, including prior authorization request counts, approval and denial rates, appeal overturn rates, and average decision times by carrier, year, service category, and code group. Service categories include inpatient and outpatient medical/surgical services, mental health and substance use disorder services, diabetes supplies and equipment, durable medical equipment, and, in more recent years, certain prescription drugs. This dataset

will be used to (a) identify high-volume and high-denial service categories and codes that define "PA-prone" services for RQ1–RQ2 and (b) provide operational benchmarks for evaluating the potential impact of AI-driven automation on approval patterns and turnaround times in RQ4.

Data sources:

- **https://www.kaggle.com/datasets/colabsss/electronic-medical-records**
- **https://www.cms.gov/data-research/statistics-trends-and-reports/medicare-claims-synthetic-public-use-files/cms-2008-2010-data-entrepreneurs-synthetic-public-use-file-de-synpuf/de10-sample-8**
- **https://data.wa.gov/api/views/fysr-7kwx/rows.json?accessType=DOWNLOAD**

Data dictionary:

Variable name	Description	Type	Source
beneficiary_id	Synthetic Medicare beneficiary ID	ID	SynPUF Beneficiary Summary
age	Age in years	Numeric	SynPUF (DOB) + Kaggle ICU
sex / gender	Sex / gender (e.g., male, female)	Categorical	SynPUF, Kaggle ICU
race_ethnicity	Race / ethnicity	Categorical	SynPUF, Kaggle ICU
HMO_months	Months enrolled in HMO	Numeric	SynPUF
diabetes_flag	Diabetes indicator	Binary	SynPUF chronic condition flag
CHF_flag	Congestive heart failure indicator	Binary	SynPUF chronic condition flag
CKD_flag	Chronic kidney disease indicator	Binary	SynPUF chronic condition flag
COPD_flag	COPD indicator	Binary	SynPUF chronic condition flag
cancer_flag	Cancer indicator	Binary	SynPUF chronic condition flag

medreimb_IP	Annual inpatient reimbursement	Numeric	SynPUF
medreimb_OP	Annual outpatient reimbursement	Numeric	SynPUF
medreimb_CAR	Annual carrier (physician) reimbursement	Numeric	SynPUF
total_annual_cost	Sum of IP, OP, and CAR reimbursements	Numeric (derived)	SynPUF
PA_prone_user	Any claim with PA-prone code in year	Binary (label)	SynPUF claims + OIC code list
high_cost_user	Total cost above high-cost threshold (e.g., top decile)	Binary (label)	Derived from total_annual_cost
apacheadmissiondx	ICU admission diagnosis (short text)	Text / categorical	Kaggle ICU
unittype	ICU type (e.g., Neuro ICU, Med-Surg ICU, CTICU)	Categorical	Kaggle ICU
unitadmitsource	ICU admission source (ED, OR, floor, direct admit)	Categorical	Kaggle ICU
LOS_ICU	ICU length of stay (approx.)	Numeric (derived)	Kaggle ICU
hospitaldischargelocation	Discharge location (home, SNF, floor, SDU, etc.)	Categorical	Kaggle ICU
hospitaldischargestatus	Alive vs deceased at discharge	Binary	Kaggle ICU
clinical_text	Free-text summary of stay	Text	Kaggle ICU
search_query	Short label of note context (e.g., “heart disease diagnosis”)	Text / categorical	Kaggle ICU
carrier	Anonymous carrier ID (Carrier A, B, etc.)	Categorical	WA OIC PA dataset
year	Reporting year	Numeric	WA OIC PA dataset
service_category	Service category (e.g., inpatient med/surg, DME, drugs)	Categorical	WA OIC PA dataset
service_code	Specific code (HCPCS/CPT/revenue/GPI/NDC)	Categorical	WA OIC PA dataset

total_PA_requests	Number of PA requests for carrier–category–code	Numeric	WA OIC PA dataset
approval_rate	Proportion of PA requests approved	Numeric (0–1)	WA OIC PA dataset
mean_decision_time_standard	Mean hours to decision for standard PA cases	Numeric	WA OIC PA dataset

Data Preprocessing

- Handling missing values
- Encoding categorical variables
- Text cleaning for NLP
- Feature scaling

Git Link:

KavyaKShetty5/PriorAuthAutomation: Research on how AI can be applied to improve the efficiency, accuracy, and scalability of the prior authorization process.

Analytic approach

- Analytical Approach
  - RQ1: Factor Analysis
    - Logistic regression
    - Hypothesis testing
    - Feature importance analysis
  - RQ2: Prediction Models
    - Logistic Regression

        - Random Forest
        - XGBoost

    - RQ3: NLP Techniques
        - TF-IDF vectorization
        - Named Entity Recognition (NER)
        - BERT (optional advanced model)

    - RQ4: Efficiency Analysis
        - t-test for comparing means
        - Before vs after automation comparison

- Evaluation Metrics
    - Classification Metrics
        - Accuracy
        - Precision
        - Recall
        - F1 Score
        - ROC-AUC
    - Operational Metrics
        - Turnaround Time Reduction (%)
        - Approval Rate Improvement
        - Cost Savings

Proposed AI Architecture

System Workflow

- Data Input (EMR systems)
- Data Preprocessing
- Feature Engineering
- Machine Learning Model
- NLP Module
- Decision Support Output

Results

The AI-based system is expected to:

- Achieve high prediction accuracy (>80%)
- Reduce turnaround time significantly
- Improve approval rates
- Enhance decision-making efficiency

Recommendation and application

Target Users

- Hospitals
- Insurance providers

- Healthcare IT companies

Applications

- EMR integration
- Revenue cycle management
- Clinical decision support systems

Limitations

Data availability constraints and generalizability across healthcare systems remain challenges *(Mello et al., 2026)*. Model interpretability also poses risks, particularly in stakeholder-facing contexts *(Lenert et al., 2023)*.

Conclusion

This study demonstrates the potential of AI-driven solutions to improve the prior authorization process in healthcare. By leveraging machine learning and NLP, the proposed system enhances efficiency, reduces delays, and improves patient outcomes.

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